

11 - Group 7

21/12/24

Group 7 elements

→ F, Cl, Br, I

→ also called 'halogens'

→ all exist as diatomic molecules (X_2)

Physical properties

⇒ atomic radius

increases down the group

⇒ melting point

increases down the group

⇒ volatility [ease to vaporize]

decreases down the group because number of electrons increases, thus strength of van der Waal's forces increases, thus boiling point increases as well as stronger ID-ID forces which decreases volatility

⇒ colors

F → pale yellow gas

Cl → yellow-green gas

Br → orange-brown liquid

I → grey-black solid

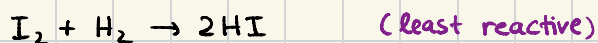
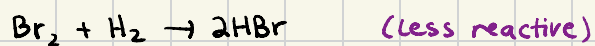
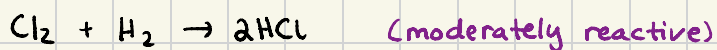
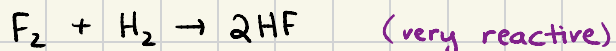
Chemical properties

* halogens act as oxidising agents (gain electrons)

* fluorine is the strongest oxidising agent, iodine is the weakest ($F_2 > Cl_2 > Br_2 > I_2$)

* down the group, reactivity ↓ because halogens get larger, thus their ability to gain electrons weakens

Reactions with hydrogen



reactivity decreases
down the group

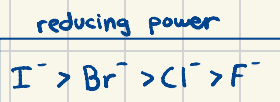
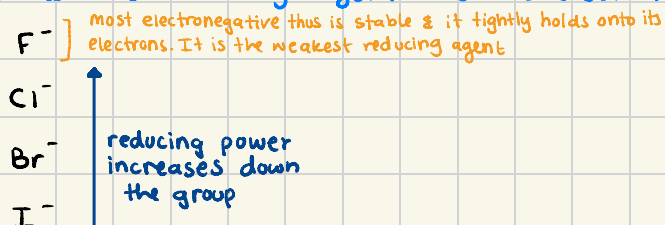
⇒ thermal stability decreases down the group (HF is most stable & HI is least stable)

↳ because as size of halogen increases, the bond between the halogen & hydrogen weakens

★ Halide ions are the ions formed when halogens gain an electron ★

Halide ions

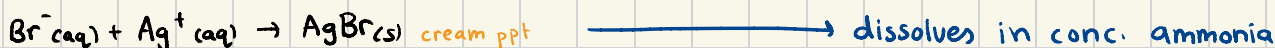
* act as reducing agents (lose electrons)



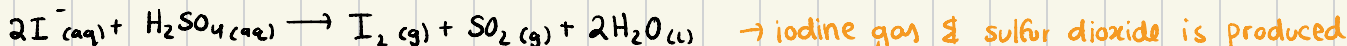
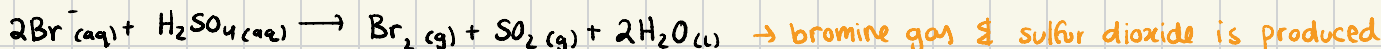
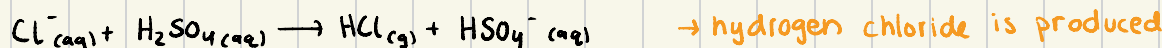
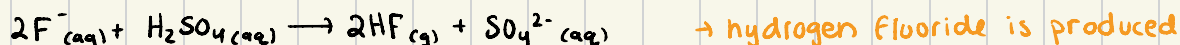
Reactions of halide ions

with aqueous silver ions

followed by aqueous ammonia



with conc. sulfuric acid



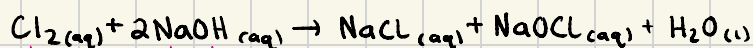
Reactions of chlorine

chlorine gets oxidised & reduced

$\rightarrow$ chlorine reacts with cold and hot aqueous NaOH in a disproportionation reaction

$\rightarrow$ chlorine is also used in water purification

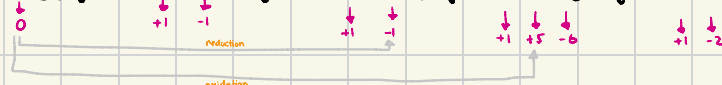
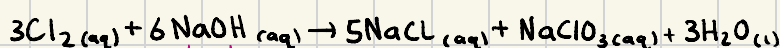
with cold NaOH



i) chlorine is oxidised to form chlorate ions (ClO_3^-)

ii) chlorine is reduced to form chloride ions (Cl^-)

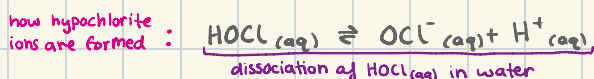
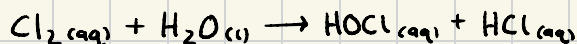
with hot NaOH



i) chlorine is oxidised to form hypochlorite ions (OCl^-)

ii) chlorine is reduced to form chloride ions (Cl^-)

with water $\rightarrow$ for water purification



produces active species like hypochlorous acid ($HOCl$) & hypochlorite ions (OCl^-), which are strong oxidisers that kill a bacteria by oxidising its cell walls & enzymes